

SEMM Primer - Calculus

Miguel Ángel Gómez
(miguel.gomez.f@berkeley.edu)

University of California, Berkeley

August 2026

Contents

- ① Introduction
- ② Integration by parts
- ③ Leibnitz rule
- ④ Numerical integration

Contents

- ① Introduction
- ② Integration by parts
- ③ Leibnitz rule
- ④ Numerical integration

Introduction

Some examples where we use integration

In **Structural Analysis**:

- Moment in a beam

$$M(x) = \int V(x) dx$$

- Resultant force and centroid of a distributed load

$$F = \int_0^L w(x) dx \quad \bar{x} = \frac{\int_0^L x w(x) dx}{\int_0^L w(x) dx}$$

- Deflections on a beam

$$v(x) = \iint \frac{M(x)}{EI(x)} dx$$

Introduction

Some examples where we use integration

In **Structural Dynamics**:

- Kinematics of a particle

$$v(t) = \int_0^t a(t) dt \quad x(t) = \int_0^t v(t) dt$$

- Kinetic and strain energy terms

$$T = \frac{1}{2} \int_V \rho v^2 dV \quad U = \frac{1}{2} \int_V \sigma \epsilon dV$$

- Integration of the equation of motion

$$x(t) = x_0 + \int_{t_0}^t F(x(t), t) dt$$

Introduction

Some examples where we use integration

And so many other topics...

- Finite Elements
- Mechanics of Solids
- Reinforced Concrete
- ...

Introduction

This will be a refresher/summary on some useful things you might have heard of before:

- Integration by parts
- Leibnitz rule
- Numerical integration

Contents

- ① Introduction
- ② Integration by parts
- ③ Leibnitz rule
- ④ Numerical integration

Integration by parts

Consider the single-fold integral over the interval $[a, b]$:

$$I = \int_a^b f(x) \, dx$$

If $F(x)$ is the integral of $f(x)$, meaning $F'(x) = f(x)$, then $I = F(b) - F(a)$.
One useful formula for integration is:

$$\int_a^b u(x)v'(x) \, dx = [u(x)v(x)]_a^b - \int_a^b u'(x)v(x) \, dx .$$

Integration by parts

A useful way to remember this formula

That seems a bit too long to remember... However, there's a little trick you can use. We all remember the product rule:

$$(uv)' = u'v + uv'$$

Rearrange:

$$uv' = (uv)' - u'v$$

Now, just integrate in the interval:

$$\int_a^b uv' \, dx = [uv]_a^b - \int_a^b u'v \, dx$$

Integration by parts

Example

A small example on how to use this formula. Let's solve:

$$I(x) = \int_0^x y^2 \exp(-y) dy$$

It's convenient to use:

$$u = y^2$$

$$du = 2y dy$$

$$dv = \exp(-y) dy$$

$$v = -\exp(-y)$$

Integration by parts

Example

$$\begin{aligned}u &= \underline{y^2} & dv &= \exp(-y) \cdot dy \\ \underline{du} &= 2y \cdot dy & v &= \underline{-\exp(-y)}\end{aligned}$$

$$\begin{aligned}\Rightarrow I(x) &= \underline{y^2} \cdot (\underline{-\exp(-y)}) \Big|_0^x - \int_0^x \underline{\exp(-y)} \cdot \underline{2y} \cdot dy \\ &= -x^2 \cdot \exp(-x) + 2 \int_0^x y \cdot \exp(-y) \cdot dy\end{aligned}$$

We apply ibp again on $\int_0^x y \cdot \exp(-y) \cdot dy$ with :

$$\begin{aligned}u &= y & dv &= \exp(-y) \cdot dy \\ du &= dy & v &= -\exp(-y)\end{aligned}$$

Integration by parts

Example

$$\begin{aligned}u &= y & dv &= \exp(-y) \cdot dy \\ du &= dy & v &= -\exp(-y)\end{aligned}$$

$$\begin{aligned}\Rightarrow \int_0^x y \cdot \exp(-y) \cdot dy &= -y \cdot \exp(-y) \Big|_0^x - \int_0^x -\exp(-y) \cdot dy \\ &= -x \cdot \exp(-x) - \exp(-y) \Big|_0^x \\ &= -x \cdot \exp(-x) - (\exp(-x) - 1)\end{aligned}$$

$$\begin{aligned}\Rightarrow \mathbb{I}(x) &= -x^2 \cdot \exp(-x) + 2(-x \exp(-x) - \exp(-x) + 1) \\ &= 2 - \exp(-x) (x^2 + 2x + 2)\end{aligned}$$

Integration by parts

Example

Putting it all together and reorganizing terms yields:

$$I(x) = 2 - \exp(-x)(x^2 + 2x + 2)$$

Contents

- ① Introduction
- ② Integration by parts
- ③ Leibnitz rule**
- ④ Numerical integration

Leibnitz rule

Sometimes, we want to take the derivative of an integral, in which the limits of integration include the variable we want to take the derivative with respect to (i.e., varying boundaries). Leibnitz rule aids with this kind of computation:

$$\frac{d}{d\alpha} \int_{h(\alpha)}^{g(\alpha)} f(x, \alpha) dx = f(g(\alpha), \alpha) \frac{dg}{d\alpha} - f(h(\alpha), \alpha) \frac{dh}{d\alpha} + \int_{h(\alpha)}^{g(\alpha)} \frac{\partial f}{\partial \alpha}(x, \alpha) dx$$

Leibnitz rule

Example

Here's a simple direct application example. Compute $\frac{d}{dx}(I(x))$, where:

$$I(x) = \int_0^{x^2} t e^{xt} dt$$

We note that, with the notation introduced before, we have:

$$\begin{array}{ll} \alpha = x & x = t \\ h(x) = 0 & g(x) = x^2 \\ f(t, x) = t e^{xt} & \end{array}$$

So, direct application of Leibnitz rule yields:

Leibnitz rule

Example

$$\begin{aligned} \alpha &= x & x &= t \\ h(x) &= 0 & g(x) &= x^2 \\ f(t, x) &= t \cdot e^{xt} \end{aligned}$$

$$\frac{d}{dx} \left(\int_{h(x)}^{g(x)} t \cdot e^{xt} \cdot dt \right) = f(g(x), x) \cdot \frac{dg}{dx} - f(h(x), x) \cdot \frac{dh}{dx} + \int_{h(x)}^{g(x)} \frac{\partial f}{\partial x}(t, x) \cdot dt$$

Diagram annotations for the Leibnitz rule formula:

- Blue arrow from $t = x^2$ to $g(x)$
- Blue arrow from $2x$ to $\frac{dg}{dx}$
- Green arrow from $t = 0$ to $h(x)$
- Green arrow from 0 to $\frac{dh}{dx}$
- Pink arrow from x^2 to $g(x)$
- Pink arrow from 0 to $h(x)$

$$\Rightarrow \frac{d}{dx} (I(x)) = x^2 \cdot e^{x^3} \cdot 2x + \int_0^{x^2} t^2 \cdot e^{xt} \cdot dt$$

Leibnitz rule

Example

$$\frac{dI}{dx} = 2x^3 e^{x^3} + \int_0^{x^2} t^2 e^{xt} dt$$

What do we do with

$$\int_0^{x^2} t^2 e^{xt} dt$$

Contents

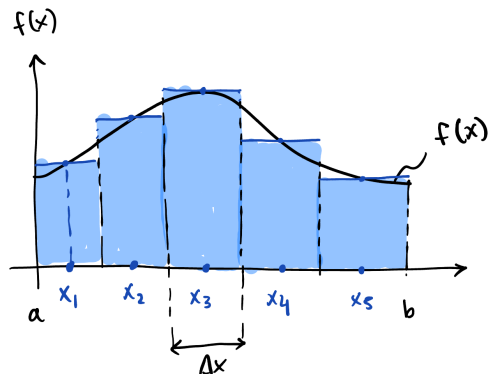
- ① Introduction
- ② Integration by parts
- ③ Leibnitz rule
- ④ Numerical integration

Numerical integration

Recall that an integral is simply computing the area under a curve. Therefore, we can define a numerical integration scheme that divides the integration interval $[a, b]$ into n equispaced discrete points, with spacing $\Delta x = (b - a)/n$ and, then:

$$I \approx \sum_{i=1}^n f(x_i) \Delta x$$

Numerical integration



More sophisticated methods locate the x_i values more efficiently, so that fewer points are required to achieve some desired accuracy (e.g., Gauss quadrature).

SEMM Primer - Calculus

Miguel Ángel Gómez
(miguel.gomez.f@berkeley.edu)

University of California, Berkeley

August 2026